Network Modelling Assignment 2

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Contents

1 Task 1: Geometrically Weighted Terms in ERGMs

```
# stocnet suite
library(manynet)
library(netrics)   # degree/density functions moved here in manynet 2.0.0
library(autograph)

# sna suite
library(network)
library(ergm)

# plotting and presentation
library(ggplot2)
library(patchwork)
library(texreg)
library(broom)
library(dplyr)
library(tidyr)

set.seed(42)
DATA <- "../Glasgow/"
```

2 Data Preparation

```r
# Read adjacency matrices with manynet
friendship <- read_matrix(paste0(DATA, "f1.csv"), header = FALSE)

# Read node attributes
attributes <- read.csv(paste0(DATA, "demographic.csv"), header = TRUE)
alcohol    <- read.csv(paste0(DATA, "alcohol.csv"),     header = TRUE)
attributes$alcohol <- alcohol$Wave.1

# Basic descriptives using manynet / netrics
nvertices <- net_nodes(friendship)
nvertices
```

```
## [1] 129
```

```r
nedges <- net_ties(friendship)
nedges
```

```
## [1] 448
```

```r
density <- net_by_density(friendship)   # moved to netrics in manynet 2.0.0
density
```

```
## [1] 0.0271
```

```r
outdegree <- node_by_outdegree(friendship, normalized = FALSE)
mean(outdegree)
```

```
## [1] 3.472868
```

```r
sd(outdegree)
```

```
## [1] 1.630172
```

```r
indegree <- node_by_indegree(friendship, normalized = FALSE)
mean(indegree)
```

```
## [1] 3.472868
```

```r
sd(indegree)
```

```
## [1] 2.267562
```

```r
plot(outdegree) + plot(indegree)
```

```r
# Add a name column so join_nodes can match by node index
attributes$name <- 1:nvertices

# Create network object for ERGM estimation (sna suite)
friendship_sna <- friendship |>
  mutate_nodes(name = 1:nvertices) |>
  join_nodes(attributes) |>
  as_network()

friendship_sna
```

```
##  Network attributes:
##   vertices = 129
##   directed = TRUE
```

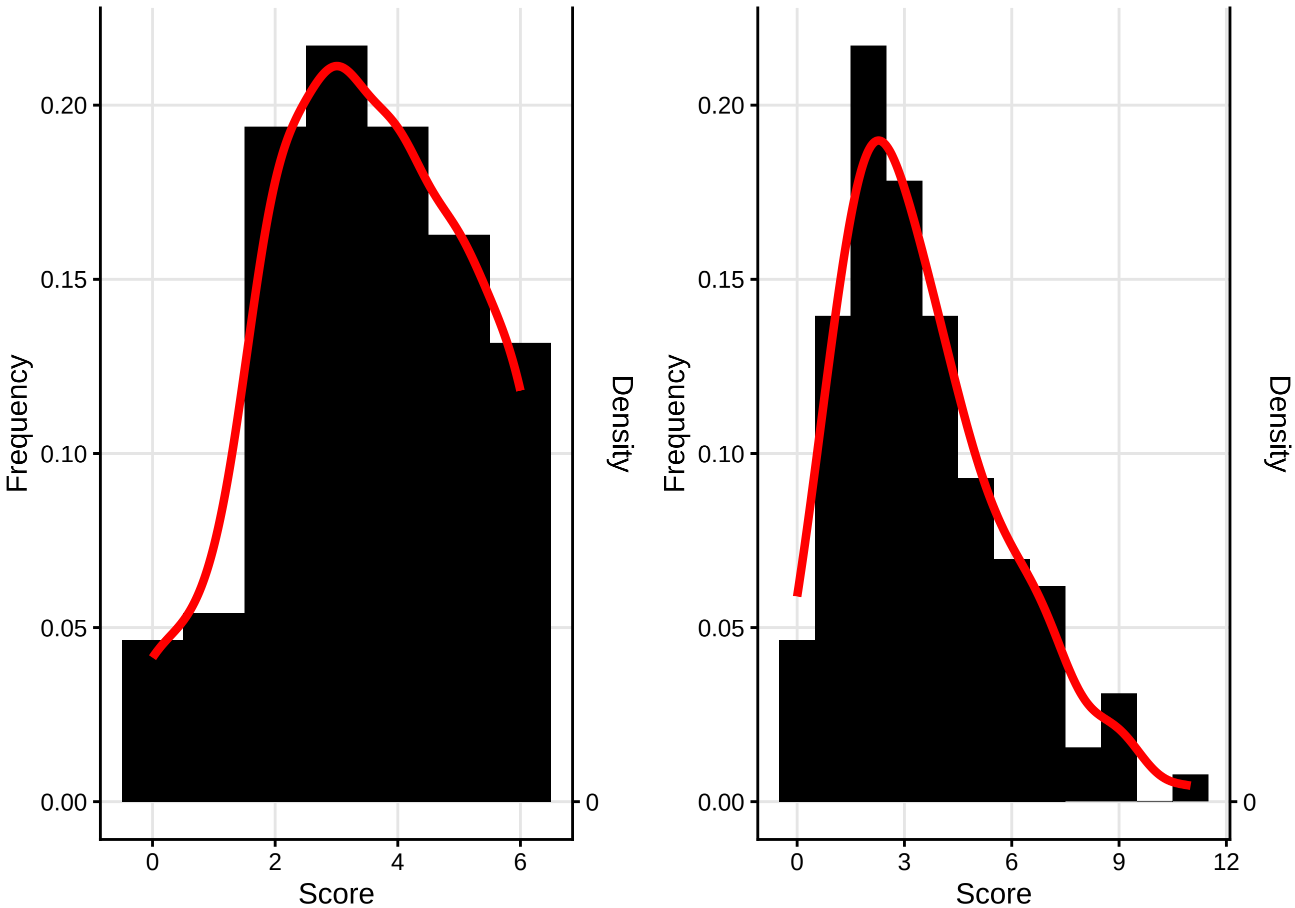

Figure 1: Out-degree and in-degree distributions of the Glasgow Wave 1 network.

```
##   hyper = FALSE
##   loops = FALSE
##   multiple = FALSE
##   bipartite = FALSE
##   total edges= 448
##     missing edges= 0
##     non-missing edges= 448
##
##  Vertex attribute names:
##     age alcohol gender vertex.names
##
## No edge attributes
```

---

3 Part (1): Change Statistic of `gwodegree`

3.1 (i) Analytical Derivation

The GWD network statistic is defined as:

$$s_{\text{GWD}}(x,\, \alpha) \;=\; e^{\alpha} \sum_{k=1}^{n} \Big\{ 1 - (1 - e^{-\alpha})^{k} \Big\}\, D_k^{\text{out}}(x)$$

where $D_k^{\text{out}}(x)$ is the number of nodes with exactly out-degree k.

Adding a directed edge to a sender node with current out-degree k:

- D_k^{out} decreases by 1 (the node leaves degree class k).
- D_{k+1}^{out} increases by 1 (the node enters degree class $k+1$).

The change statistic is therefore:

$$\begin{aligned}
\delta_{\text{GWD}}(k, \alpha) &= s_{\text{GWD}}(x + \text{edge},\, \alpha) - s_{\text{GWD}}(x,\, \alpha) \\
&= e^{\alpha} \Big[\big\{ 1 - (1 - e^{-\alpha})^{k+1} \big\} - \big\{ 1 - (1 - e^{-\alpha})^{k} \big\} \Big] \\
&= e^{\alpha} \Big[(1 - e^{-\alpha})^{k} - (1 - e^{-\alpha})^{k+1} \Big] \\
&= e^{\alpha}\, (1 - e^{-\alpha})^{k} \; \underbrace{\big[1 - (1 - e^{-\alpha}) \big]}_{= e^{-\alpha}} = \boxed{\, (1 - e^{-\alpha})^{k} \,}
\end{aligned}$$

3.2 (ii) Plots of the Change Statistic

```r
delta_GWD <- function(k, alpha) (1 - exp(-alpha))^k

alpha_vals <- c(-1, -0.5, 0, 0.1, 0.2, 0.3, 0.5, log(2), 1, 2, 3)
alpha_labs <- c("-1", "-0.5", "0", "0.1", "0.2", "0.3",
                "0.5", "ln2",  "1", "2",   "3")

# Use the range of observed out-degrees
k_max   <- max(outdegree)
k_range <- 0:k_max
```

```r
df_plot <- do.call(rbind, lapply(seq_along(alpha_vals), function(i) {
  data.frame(
    k     = k_range,
    delta = delta_GWD(k_range, alpha_vals[i]),
    alpha = factor(alpha_labs[i], levels = alpha_labs),
    group = ifelse(alpha_vals[i] < 0, "negative", "non-negative")
  )
}))
```

3.2.1 Negative decay values (one figure per value)

```r
df_sub <- df_plot |> filter(alpha == "-1")
ggplot(df_sub, aes(x = k, y = delta)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_line(colour = "steelblue", linewidth = 0.9) +
  geom_point(colour = "steelblue", size = 2.5) +
  labs(title    = expression("Change statistic," ~ alpha == -1),
       subtitle = "Negative decay value",
       x = "Current out-degree k",
       y = expression(delta[GWD](k, alpha))) +
  theme_bw(base_size = 13)
```

```r
df_sub <- df_plot |> filter(alpha == "-0.5")
ggplot(df_sub, aes(x = k, y = delta)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_line(colour = "tomato", linewidth = 0.9) +
  geom_point(colour = "tomato", size = 2.5) +
  labs(title    = expression("Change statistic," ~ alpha == -0.5),
       subtitle = "Negative decay value",
       x = "Current out-degree k",
       y = expression(delta[GWD](k, alpha))) +
  theme_bw(base_size = 13)
```

3.2.2 Non-negative decay values (one combined figure)

```r
df_nonneg <- df_plot |> filter(group == "non-negative")

ggplot(df_nonneg, aes(x = k, y = delta, colour = alpha, linetype = alpha)) +
  geom_line(linewidth = 0.9) +
  geom_point(size = 2.5) +
  scale_colour_brewer(palette = "Set1", name = expression(alpha)) +
  scale_linetype_discrete(name = expression(alpha)) +
  labs(title = "Change statistic - non-negative decay values",
       x = "Current out-degree k",
       y = expression(delta[GWD](k, alpha))) +
  theme_bw(base_size = 13)
```

3.3 (iii) Main Takeaways

The change statistic $\delta_{\text{GWD}}(k, \alpha) = (1 - e^{-\alpha})^k$ reveals how the decay parameter α governs the marginal contribution of an additional out-edge to an already-popular sender node.

Non-negative α. When $\alpha > 0$ the base $(1 - e^{-\alpha}) \in (0, 1)$, so the change statistic is always positive and decreases *monotonically* with k. Adding an edge from a node with $k = 0$ always contributes the maximum

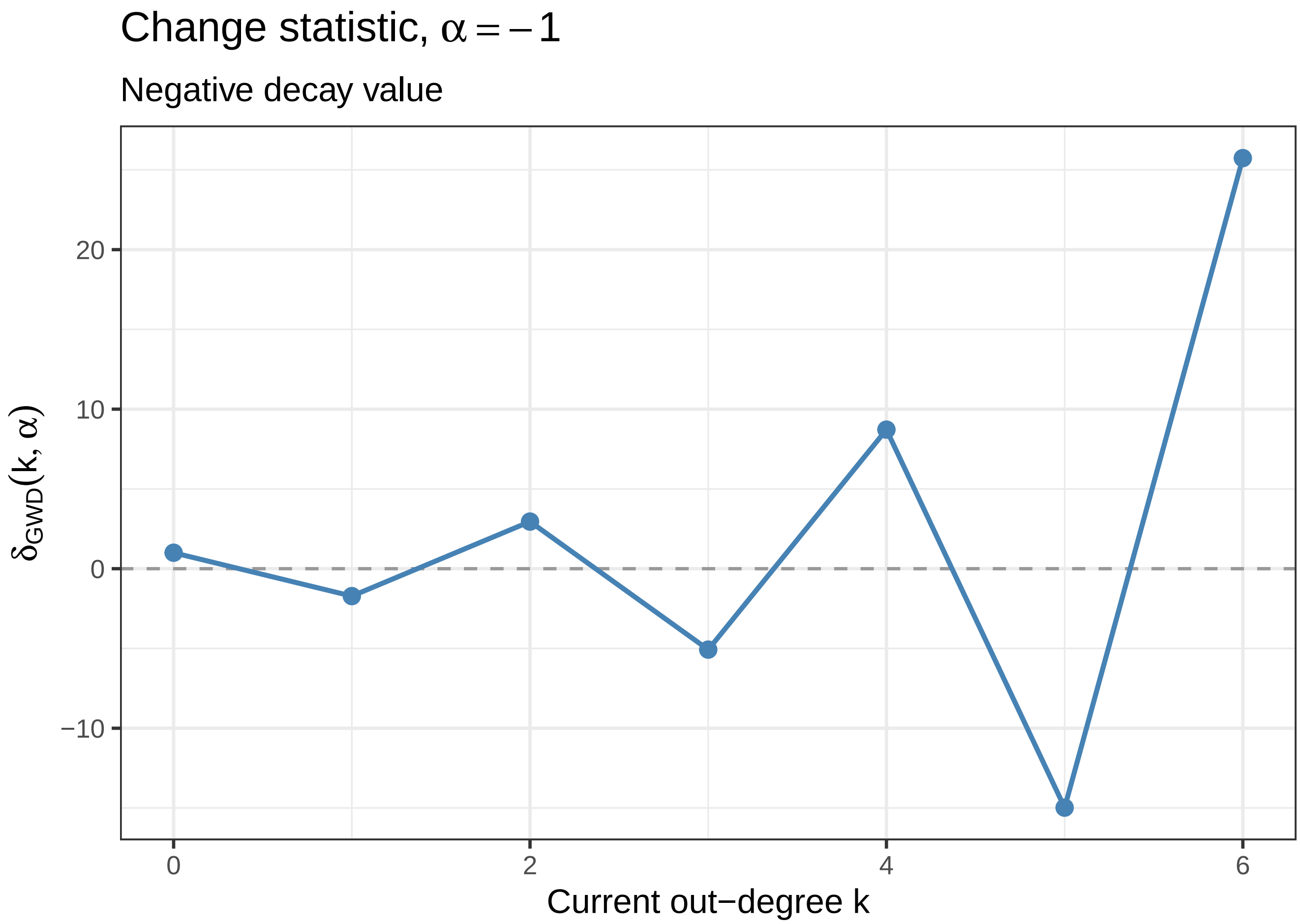

Figure 2: Change statistic for alpha = -1.

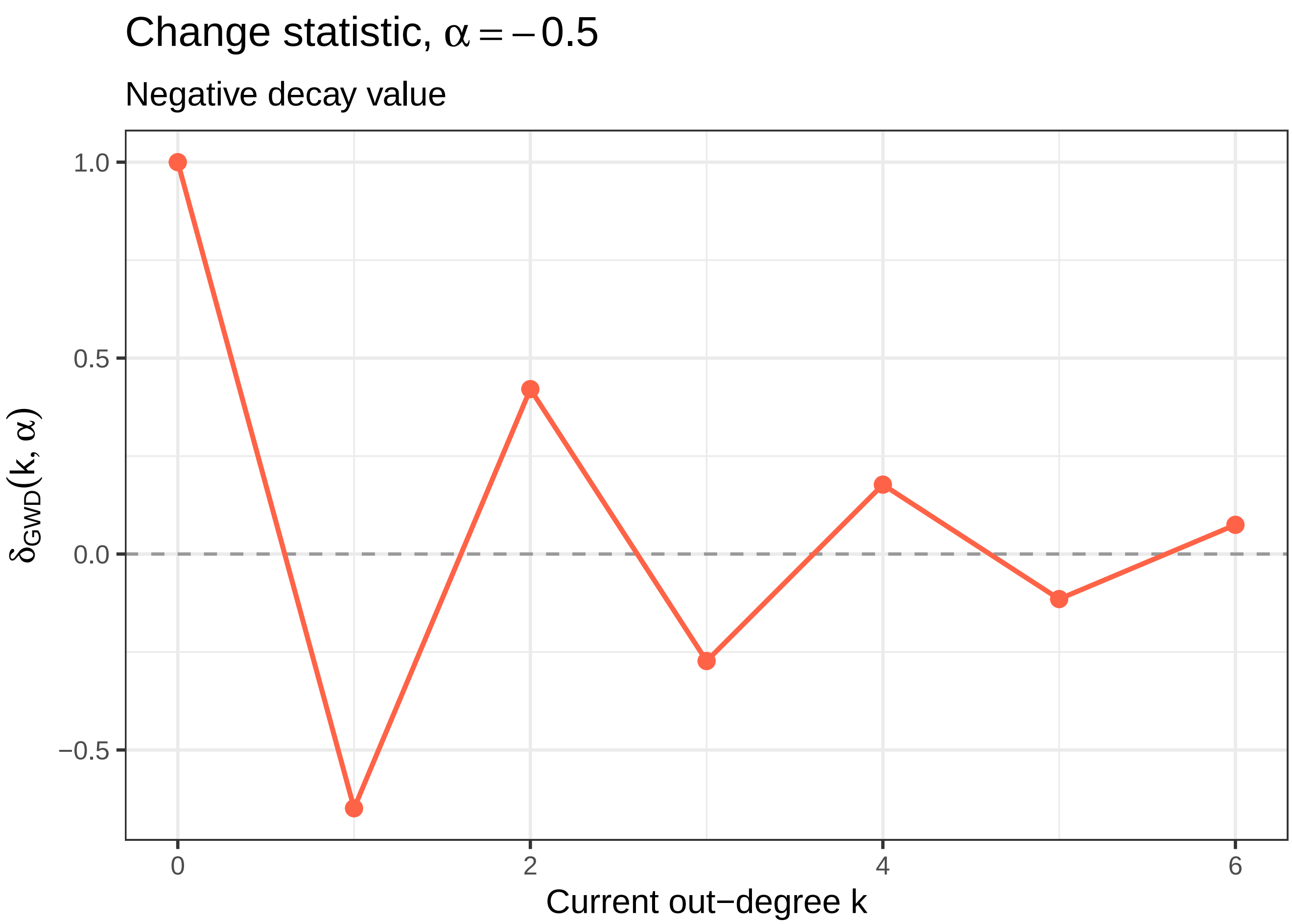

Figure 3: Change statistic for alpha = -0.5.

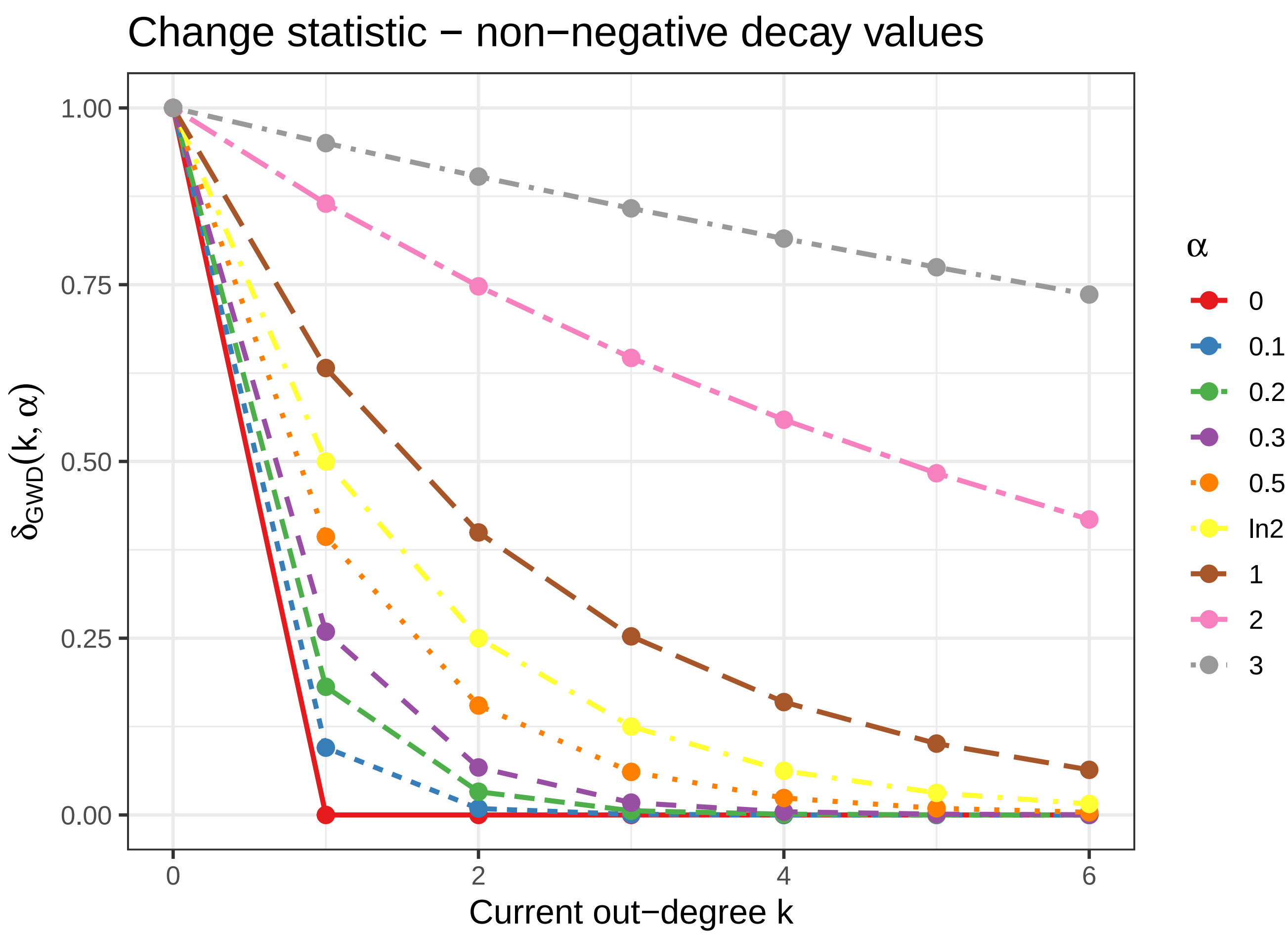

Figure 4: Change statistic for all non-negative decay values. Different colours and linetypes distinguish alpha values.

value of 1, regardless of α, while adding from a high-degree node contributes progressively less. The rate of this decay is entirely determined by α:

- *Small* α (e.g. $\alpha = 0.1$): the base is close to 0, so δ drops almost immediately to near-zero after $k = 0$. Only ties sent by near-isolated nodes influence the statistic; hubs are practically invisible.
- *Large* α (e.g. $\alpha = 2$ or $\alpha = 3$): the base is close to 1, so δ remains substantial even at high k. Both low- and high-degree senders contribute almost equally, and the statistic is sensitive to the entire out-degree spectrum.
- $\alpha = \ln 2$: the base equals $\frac{1}{2}$, giving a clean geometric halving with each additional out-edge — a natural "neutral" reference.

When $\alpha = 0$ the change statistic is 0 for all $k > 0$: adding an edge from any already-active sender has no effect on the statistic whatsoever.

Negative α. The base $(1 - e^{-\alpha}) < 0$, so δ *alternates in sign* (positive for even k, negative for odd k) and grows in absolute magnitude. This has no natural interpretation in terms of a degree distribution; the statistic oscillates rather than capturing a smooth preferential-attachment or equalisation mechanism.

Summary. The decay parameter acts as a "bandwidth" on the out-degree axis. Larger α extends the effective range over which additional ties are counted, making the statistic more sensitive to hubs; smaller positive α concentrates the statistic on low-degree nodes. Negative α produces counter-intuitive alternating behaviour and should be avoided in substantive applications.

3.4 (iv) Choosing α for an Activity Effect

An *activity* effect means that nodes who are already highly active (high out-degree) are *more* likely to send further nominations. The goal is for the `gwodegree` statistic to be *driven by* high-out-degree nodes, so that its coefficient θ_{GWD} can capture this self-reinforcing dynamic.

The change statistic $(1 - e^{-\alpha})^k$ is smallest for large k when α is small, and approaches 1 for all k as $\alpha \to \infty$. For `gwodegree` to be sensitive to hubs we need δ to remain meaningfully large at the high end of the observed out-degree range.

In the Glasgow Wave 1 network the maximum out-degree is 6. At $\alpha = 0.3$: $\delta(6,\, 0.3) = (1 - e^{-0.3})^6 \approx 0.0003$ – the sixth tie is essentially invisible. At $\alpha = 1$: $\delta(6,\, 1) \approx 0.063$; at $\alpha = 2$: ≈ 0.46; at $\alpha = 3$: ≈ 0.78.

Therefore, to model an activity effect where *high out-degree nodes drive the statistic*, α **should be chosen large** (at least $\alpha = 1$, preferably $\alpha = 2$ or $\alpha = 3$ for this network). With a large, positive θ_{GWD} and a large α, the model assigns higher probability to networks in which already-active senders continue to accumulate further nominations — a "rich-get-richer" / Matthew effect in out-degree.

Negative α must be avoided: the alternating-sign behaviour of δ makes θ_{GWD} uninterpretable and estimation prone to degeneracy.

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4 Part (2): Effect of the `gwodegree` Coefficient on the Out-Degree Distribution

4.1 (i) M1: Edges-Only Reference Model

```r
M1 <- ergm(friendship_sna ~ edges,
           control = control.ergm(seed = 42))
summary(M1)
```

```
## Call:
## ergm(formula = friendship_sna ~ edges, control = control.ergm(seed = 42))
##
```

```
## Maximum Likelihood Results:
##
##       Estimate Std. Error MCMC % z value Pr(>|z|)
## edges  -3.5795     0.0479      0  -74.73   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
##  Residual Deviance:  4116  on 16511  degrees of freedom
##
## AIC: 4118  BIC: 4125  (Smaller is better. MC Std. Err. = 0)
```

4.2 (ii) M2: Fix Decay = 0.3, Vary gwodegree Coefficient

The gwodegree coefficient is fixed via offset() while only the edges parameter is re-estimated, ensuring the expected network density matches the observed density.

```
M2a <- ergm(
  friendship_sna ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
  offset.coef = c(-2),
  control     = control.ergm(seed = 42)
)
summary(M2a)
```

```
## Call:
## ergm(formula = friendship_sna ~ edges + offset(gwodegree(0.3,
##     fixed = TRUE)), offset.coef = c(-2), control = control.ergm(seed = 42))
##
## Monte Carlo Maximum Likelihood Results:
##
##                           Estimate Std. Error MCMC % z value Pr(>|z|)
## edges                     -3.36557    0.04026      1  -83.59   <1e-04 ***
## offset(gwodeg.fixed.0.3) -2.00000    0.00000      0    -Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
##  Residual Deviance:  4116  on 16510  degrees of freedom
##
## AIC: 4118  BIC: 4126  (Smaller is better. MC Std. Err. = 1.125)
##
##  The following terms are fixed by offset and are not estimated:
##    `offset(gwodeg.fixed.0.3)`
##
```

```
M2b <- ergm(
  friendship_sna ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
  offset.coef = c(2),
  control     = control.ergm(seed = 42)
)
summary(M2b)
```

```
## Call:
## ergm(formula = friendship_sna ~ edges + offset(gwodegree(0.3,
##     fixed = TRUE)), offset.coef = c(2), control = control.ergm(seed = 42))
##
```

```
## Monte Carlo Maximum Likelihood Results:
##
##                          Estimate Std. Error MCMC % z value Pr(>|z|)
## edges                    -3.69965    0.05089      1   -72.7   <1e-04 ***
## offset(gwodeg.fixed.0.3)  2.00000    0.00000      0     Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
##  Residual Deviance:  4114  on 16510  degrees of freedom
##
## AIC: 4116  BIC: 4123  (Smaller is better. MC Std. Err. = 0.6352)
##
##  The following terms are fixed by offset and are not estimated:
##    `offset(gwodeg.fixed.0.3)`
##
```

Model coefficient table:

```
screenreg(list(M1, M2a, M2b),
          custom.model.names = c("M1", "M2a (theta=-2)", "M2b (theta=+2)"),
          caption = "Estimated edges coefficients. gwodegree coefficients are fixed via offset.")
```

==
M1 M2a (theta=-2) M2b (theta=+2) ——————————————————————- edges -3.58 ***
-3.37 *** -3.70 ***
(0.05) (0.04) (0.05)
offset(gwodeg.fixed.0.3) -2.00 *** 2.00 ***
(0.00) (0.00)
——————————————————————- AIC 4117.65 4118.05 4115.71
BIC 4125.36 4125.76 4123.42
Log Likelihood -2057.82 -2058.02 -2056.85
==
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

4.3 (iii) Simulate 100 Networks and Plot Out-Degree Distributions

Following the practical, we use `ergm::simulate_formula()` with the `coef` argument to fix parameter values.

```
NSIM <- 100

# simulate_formula() with coef fixes all parameter values
sim_M1  <- simulate_formula(
  friendship_sna ~ edges,
  coef     = coef(M1),
  nsim     = NSIM,
  output   = "network"
)

sim_M2a <- simulate_formula(
  friendship_sna ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
  coef     = c(coef(M2a)["edges"], -2),
  nsim     = NSIM,
  output   = "network"
)
```

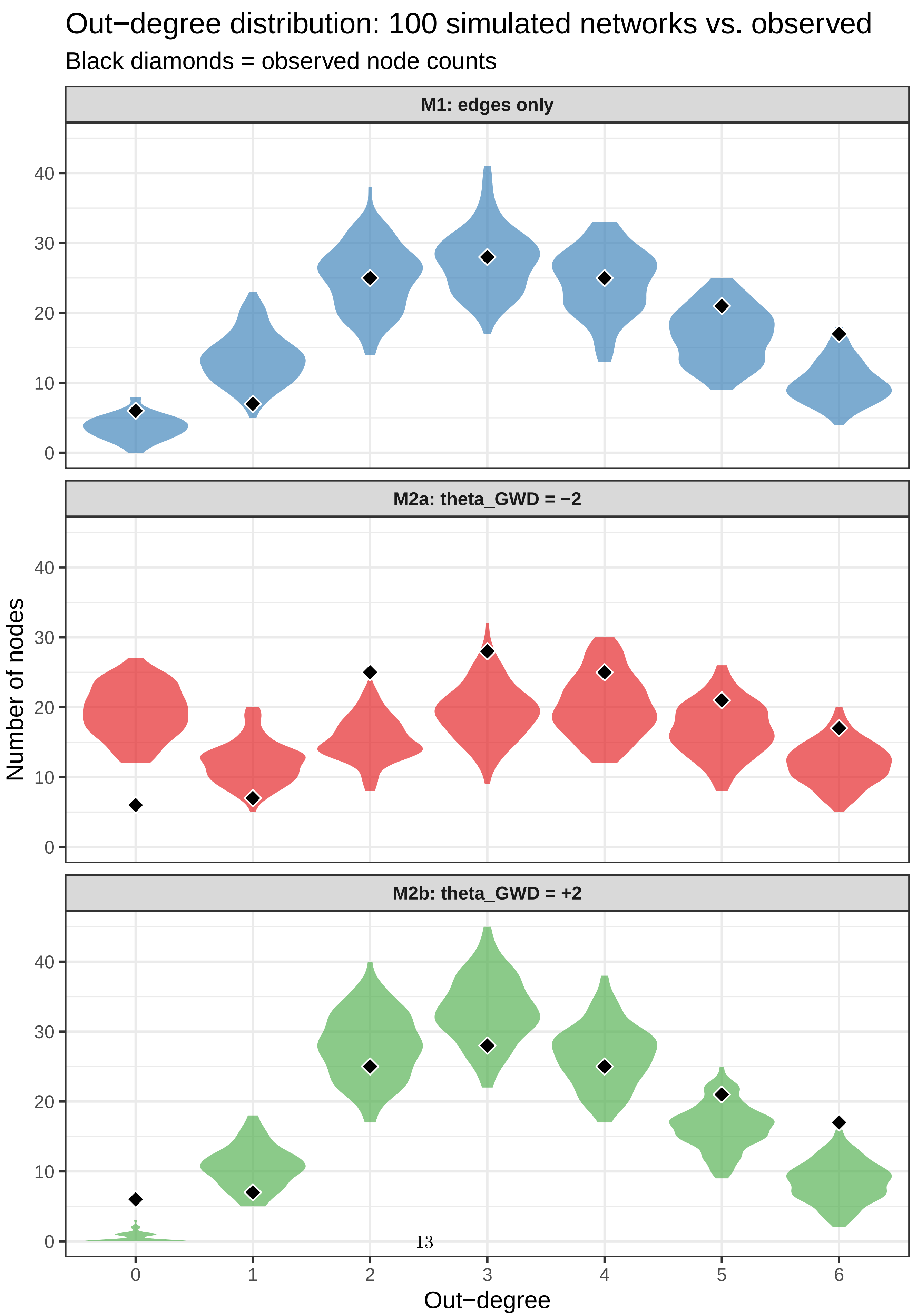
```r
sim_M2b <- simulate_formula(
  friendship_sna ~ edges + offset(gwodegree(0.3, fixed = TRUE)),
  coef    = c(coef(M2b)["edges"], 2),
  nsim    = NSIM,
  output  = "network"
)
```

```r
# Compute node count by out-degree for each simulated network
# (simulated networks are 'network' objects -> use sna::degree)
outdeg_counts <- function(sim_list, label) {
  k_max_obs <- max(outdegree)
  do.call(rbind, lapply(seq_len(NSIM), function(i) {
    degs <- sna::degree(sim_list[[i]], cmode = "outdegree")
    tab  <- tabulate(degs + 1L, nbins = k_max_obs + 1L)
    data.frame(outdegree = 0:k_max_obs, count = tab, sim = i, model = label)
  }))
}

df_sim <- bind_rows(
  outdeg_counts(sim_M1,  "M1: edges only"),
  outdeg_counts(sim_M2a, "M2a: theta_GWD = -2"),
  outdeg_counts(sim_M2b, "M2b: theta_GWD = +2")
) |>
  mutate(model = factor(model,
                        levels = c("M1: edges only",
                                   "M2a: theta_GWD = -2",
                                   "M2b: theta_GWD = +2")))

# Observed node counts by out-degree
obs_tab <- data.frame(
  outdegree = 0:max(outdegree),
  count     = tabulate(outdegree + 1L, nbins = max(outdegree) + 1L)
)
```

```r
ggplot(df_sim, aes(x = factor(outdegree), y = count, fill = model)) +
  geom_violin(scale = "width", alpha = 0.65, colour = NA) +
  geom_point(
    data        = obs_tab,
    aes(x       = factor(outdegree), y = count),
    inherit.aes = FALSE,
    shape = 23, size = 3, fill = "black", colour = "white"
  ) +
  facet_wrap(~ model, ncol = 1) +
  scale_fill_manual(
    values = c("M1: edges only"       = "#377EB8",
               "M2a: theta_GWD = -2"  = "#E41A1C",
               "M2b: theta_GWD = +2"  = "#4DAF4A")
  ) +
  labs(title    = "Out-degree distribution: 100 simulated networks vs. observed",
       subtitle = "Black diamonds = observed node counts",
       x = "Out-degree", y = "Number of nodes") +
  theme_bw(base_size = 13) +
  theme(legend.position = "none",
        strip.text      = element_text(face = "bold"))
```


Out−degree distribution: 100 simulated networks vs. observed
Black diamonds = observed node counts
M1: edges only
M2a: theta_GWD = −2
M2b: theta_GWD = +2
Number of nodes
Out−degree
0
10
20
30
40
0
1
2
3
4
5
6

4.4 (iv) Comparison and Interpretation

The three models share the same observed density by construction (the `edges` coefficient is re-estimated in M2a/M2b to compensate for the `gwodegree` offset), yet they produce strikingly different out-degree distributions.

M1 – edges only (reference). With only the `edges` term the model is a directed Bernoulli random graph in which each dyad is independently connected with a constant probability $p = \text{logistic}(\hat{\theta}_{\text{edges}})$. This produces a Binomial – and for small p approximately Poisson – out-degree distribution. In the Glasgow network ($n = 129$, density ≈ 0.027) the simulated distributions are compact, centred around 3–4, with very little mass above 6. The observed counts fall broadly within the simulated range, confirming the simple random graph as a valid baseline.

M2a ($\theta_{\mathbf{GWD}} = -2$, decay $= 0.3$). A negative `gwodegree` coefficient *penalises* the GWD statistic. Because the change statistic $(1 - e^{-0.3})^k$ is largest for $k = 0$ and drops rapidly, penalising this term discourages ties sent from low-degree nodes while being less averse to ties from high-degree ones. The result is a more *heterogeneous*, right-skewed out-degree distribution: fewer nodes remain at degree 0 or 1, and more mass appears at moderate-to-high degrees. The distribution is broader than M1 and exhibits a heavier tail.

M2b ($\theta_{\mathbf{GWD}} = +2$, decay $= 0.3$). A positive `gwodegree` coefficient *rewards* the GWD statistic. Because the reward is greatest for ties from low-degree senders, the model strongly favours nodes sending only a few ties and discourages further accumulation. The simulated distributions are markedly more *concentrated* than M1: most nodes cluster at a common moderate out-degree (2–4) with almost no nodes at 0 or at the upper tail. The variance of the out-degree distribution is substantially reduced compared to both M1 and M2a.

Comparison with a random network of the same density. The Bernoulli baseline (M1) already produces a relatively narrow near-Poisson distribution. $\theta_{\text{GWD}} < 0$ broadens and right-skews this distribution, introducing *activity heterogeneity*: some students nominate many friends while others nominate few. $\theta_{\text{GWD}} > 0$ compresses it further, enforcing *homogeneous activity* across students. In substantive terms, a positive `gwodegree` coefficient captures a tendency toward equal participation, while a negative one allows hubs to emerge among senders.